

DTMC

Chance

1. PCTL formulas are ϕ with the following syntax:

$$\begin{array}{ll} \phi ::= \mathbf{true} \mid p \mid \phi_1 \wedge \phi_2 \mid \neg\phi \mid \mathbf{P}_{\sim p}[\psi] & \text{(state formulas)} \\ \psi ::= \mathbf{X}\phi \mid \phi_1 \mathbf{U}^{\leq k} \phi_2 \mid \phi_1 \mathbf{U} \phi_2 & \text{(path formulas)} \end{array}$$

where p is an atomic proposition, and $\sim$ is one of $\leq, <, \geq, >, =$.

- (a) Give the formal semantic of PCTL, interpreted over DTMC.

Answer: Below (see also the slides on DTMC).

Let $M = (S, s_0, P, L)$ be your DTMC. A PCTL formula ϕ holds on M if ϕ holds on the initial state s_0 , denoted by $s_0 \models \phi$. We now give meaning of $s_0 \models \phi$, for any state s . See below:

$$\begin{array}{ll} s \models p & = p \in L(s) \\ s \models \neg\phi & = \text{not}(s \models \phi) \\ s \models \phi_1 \wedge \phi_2 & = s \models \phi_1 \text{ and } s \models \phi_2 \end{array}$$

For the probabilistic operator $\mathbf{P}_{\sim p}[\psi]$ the definition is as below. I only give the def. for ' $>$ ' as the $\sim$. The definition is similar for other cases of $\sim$.

$$s \models \mathbf{P}_{>q}\psi = \underbrace{P\{\omega \in Paths(s) \mid \omega \models \psi\}}_{\text{probability}} > q$$

Here, $Paths(s)$ is the set of all possible paths/executions that starts from s ; typically there are infinitely many of them. The notation $\omega \models \psi$ means that the (temporal) property ψ holds on the path ω .

Finally, the notation $P\{\omega \in Paths(s) \mid \omega \models \psi\}$ refers to the probability that such a path ω occur as the behavior of the model if we start in the state s .

The meaning of $\omega \models \psi$ is defined recursively over the syntax of ψ as follows (the def. is not complete, you can see the slides for the rest):

$$\begin{array}{ll} \omega \models \mathbf{X}\phi & = \omega[1..] \models \phi \\ \omega \models \phi_1 \mathbf{U} \phi_2 & = \omega \models \phi_2 \text{ or } (\omega \models \phi_1 \text{ and } \omega[1..] \models \phi_1 \mathbf{U} \phi_2) \end{array}$$

The notation $\omega[1..]$ refers to the (infinite) suffix of ω starting from index 1 (so, the sequence $\omega_1, \omega_2, \dots$).

- (b) Let's introduce some derived operators with the 'usual' meaning: $\vee$, $\rightarrow$, and $\mathbf{F}$ (or $\Diamond$). Propose a definition for them.

Answer: This can be defined as 'usual':

$$\begin{array}{ll} \phi_1 \vee \phi_2 & = \neg(\neg\phi_1 \wedge \neg\phi_2) \\ \phi_1 \rightarrow \phi_2 & = \neg\phi_1 \vee \phi_2 \\ \mathbf{F}\phi & = \mathbf{true} \mathbf{U} \phi \end{array}$$

- (c) Developer Peter Parker claims that $\neg\mathbf{P}_{>c}\phi$ is equivalent to saying $\mathbf{P}_{\leq c}\phi$. Is Peter right?

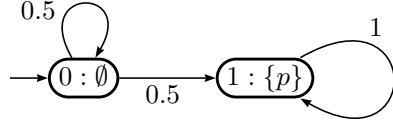
Answer: see in the slides :)

- (d) The 'usual' way to define $\mathbf{G}$ is: $\mathbf{G}\phi = \neg\mathbf{F}\neg\phi$. So, for example $\mathbf{P}_{\geq 0.8}[\mathbf{G}a]$ can be expressed as $\mathbf{P}_{\geq 0.8}[\neg\mathbf{F}\neg a]$. Unfortunately, the syntax of PCTL does not have negation over the temporal formula (we only have negation over proposition and probabilistic formulas). So, the formula " $\mathbf{P}_{\geq 0.8}[\neg\mathbf{F}\neg a]$ " is not allowed in PCTL. How do you then propose to define $\mathbf{P}_{\geq p}[\mathbf{G}\phi]$?

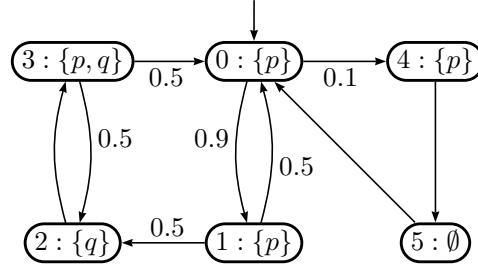
Answer: How about like this: $\mathbf{P}_{\leq p}[\mathbf{F}\neg\phi]$?

2. Explain why in PCTL (e.g. interpreted over DTMC) $\mathbf{P}_{\geq 1}[\mathbf{F}\phi]$ is not the same as CTL $\mathbf{AF}\phi$ (the former is weaker than the latter). Give an example that shows the difference. Note: $\mathbf{F}\phi$ is defined as $\text{true} \mathbf{U} \phi$.

Answer: Consider the DTMC M below. The CTL property $\mathbf{AF}p$ is not a valid property of M , because it can cycle in $S0$ forever, hence never reaching the state $S1$ where p holds. However, from PCTL perspective, the probability that M cycles in $S0$ forever would approach zero. And therefore the probability of eventually going over to $S1$ approaches 1.



3. Consider the following DTMC M modelling some system. $Prop = \{p, q\}$.



Consider the following PCTL properties to check:

- $\mathbf{P}_{\geq 0.82} [p \mathbf{U} q]$
- $\mathbf{P}_{< 0.82} [p \mathbf{U} q]$
- $\mathbf{P}_{=?} [p \mathbf{U} q]$

What do these formula say? Then describe the procedure to check them (or at least one of them) on M .

Answer:

The property $\mathbf{P}_{\geq 0.82} [p \mathbf{U} q]$ is valid on a state s if the execution that starts from s would satisfy the path formula $\psi = p \mathbf{U} q$ with the probability of at least 0.82. When the reference state s is not mentioned, we mean it to be M 's initial state. Since all executions of M starts from its initial state, this is the same as saying that M 's executions satisfy the asserted (probabilistic) property.

The formula $\mathbf{P}_{=?} [p \mathbf{U} q]$ is a quantitative query (so, not a Y/N-query), asking what the probability of "a state s " to satisfy the path property $p \mathbf{U} q$. If no reference state is mentioned, then we mean to ask that on s_{init} (well, M 's initial state).

Also notice that answering the query $\mathbf{P}_{=?} [p \mathbf{U} q]$ for every state s in M will enable us to answer the other two queries/formulas above.

Let's now focus on the last formula; let's call the inner formula ψ :

$$\mathbf{P}_{=?} [\underbrace{p \mathbf{U} q}_{\psi}]$$

Let us now calculate, for every state, the probability that it satisfies ψ . We will proceed as follows:

- (a) We need to first perform "precomputation", to identify the so-called $\mathbf{S}^{yes}$ and $\mathbf{S}^{no}$ sets. We start with $\mathbf{S}^{no}$. This is the set of states of M on which $\mathbf{P}_{\leq 0}[\psi]$ holds. From the picture you can 'see' that this set is $\{4, 5\}$, but let us see how we can calculate this systematically.

Calculate first the set E , which is all the states satisfying the (non-probabilistic) CTL formula $\mathbf{E}[p \mathbf{U} q]$. You can do this e.g. with CTL standard model-checking. We end up with $E = \{0, 1, 2, 3\}$. Any state in E has a path satisfying ψ , and hence a non-zero probability of satisfying ψ . So, $\mathbf{S}^{no}$ must be $S/E = \{4, 5\}$.

The set $\mathbf{S}^{yes}$ consists of all states on which $\mathbf{P}_{\geq 1}[\psi]$ holds. To calculate this set, we first calculate the set F of all states t that have a path τ , starting from t , leading to $\mathbf{S}^{no}$, while maintaining the property $p \wedge \neg q$ along τ (up to entering until $\mathbf{S}^{no}$). We end up with $F = \{0, 1, 4, 5\}$ ¹. By its definition/construction, states in F has a non-zero probability to satisfy $\neg\psi$. So, $S/F = \{2, 3\}$ is the set $\mathbf{P}_{\geq 1}[\psi]$ we look for²

So, to summarize we have calculated:

- $\mathbf{S}^{no} = \{4, 5\}$.
- $\mathbf{S}^{yes} = \{2, 3\}$.

- (b) We can now calculate the probability of each state in M for satisfying ψ .

We can conclude that the probability of any state in $\mathbf{S}^{no}$ for satisfying ψ is simply 0. Conversely, the probability of any state in $\mathbf{S}^{yes}$ for satisfying ψ is 1.

For any state t that do not belong to either $\mathbf{S}^{yes}$ or $\mathbf{S}^{no}$, its probability for satisfying ψ is some $0 < p < 1$. We have two such states: $\{0, 1\}$. Let x_s denote the probability that a state s satisfies ψ (which is $p \mathbf{U} q$). We will have the following relations/equations; and x_i 's are the solutions of the equations:

$$\begin{aligned} x_0 &= P_{0 \rightarrow 1} * x_1 + P_{0 \rightarrow 4} * x_4 \\ x_1 &= P_{1 \rightarrow 0} * x_0 + P_{1 \rightarrow 2} * x_2 \end{aligned}$$

where $P_{u \rightarrow v}$ denotes the probability of taking the transition to v , if we are in the state u .

Now, we know that $x_4 = 0$ and $x_2 = 1$. So the above simplify to:

$$\begin{aligned} x_0 &= P_{0 \rightarrow 1} * x_1 \\ x_1 &= P_{1 \rightarrow 0} * x_0 + P_{1 \rightarrow 2} \end{aligned}$$

Filling the the values of the P (from M 's picture) we obtain these equations:

$$\begin{aligned} x_0 &= 0.9x_1 \\ x_1 &= 0.5x_0 + 0.5 \end{aligned}$$

You can apply algebra to solve the equations. E.g. the first says x_0 is $0.9x_1$. If we fill this in for the x_0 in the 2nd equation we get:

$$x_1 = 0.5 * (0.9x_1) + 0.5$$

This leads to the solution that $x_1 \approx 0.909$. And $x_0 \approx 0.818$.

¹Notice that $\{4, 5\}$ are states from $\mathbf{S}^{no}$ itself. In general, this F will indeed always include $\mathbf{S}^{no}$.

²The MC-slides in the website of Prism, F is characterized differently, namely as the set of states where the probability of ψ is < 1 . This should be the same as saying that there is a non-zero probability of $\neg\psi$, which is how I looked at it above.

(c) Now we can answer all formulas/queries that were posed in the original question:

- $\mathbf{P}_{=?} [p \mathbf{U} q]$. On s_{init} the probability is 0.818.
- $\mathbf{P}_{\geq 0.82} [p \mathbf{U} q]$. Well, then this is **not** valid on s_{init} .
- $\mathbf{P}_{< 0.82} [p \mathbf{U} q]$. This is valid on s_{init} .

4. Consider again the DTMC M in No. 3. On which states of M the following properties hold?

- $\mathbf{P}_{\leq 0} [p \mathbf{U} \neg(p \vee q)]$.
- $\mathbf{P}_{\geq 1} [p \mathbf{U} \neg(p \vee q)]$.
- $\mathbf{P}_{\geq 0.18} [p \mathbf{U} \neg(p \vee q)]$.
- $\mathbf{P}_{< 0.18} [p \mathbf{U} \neg(p \vee q)]$.

5. Consider again the DTMC M in No. 3. Which states satisfy $\mathbf{P}_{\leq 0.5} [p \mathbf{U} \mathbf{P}_{\geq 1}[q \mathbf{U} p]]$? (Well, obviously state 5, but which other states, if any, satisfy the property?)

Answer: Let's first introduce some names to abbreviate:

$$\mathbf{P}_{\leq 0.5} [p \mathbf{U} \overbrace{\mathbf{P}_{\geq 1} [q \mathbf{U} p]}^{\chi}]$$

ψ

We will first calculate which states satisfy $\chi = \mathbf{P}_{\geq 1}[\psi]$. We proceed with the checking procedure as in the examples in (3):

- We calculate the set E of states satisfying $\mathbf{E}[q \mathbf{U} p]$, and obtain $E = \{0, 1, 2, 3, 4\}$. So, $\mathbf{S}^{no}$ for ψ is $\{5\}$ (the complement of E).
- Calculate the set F consisting states with an outgoing path whose states satisfy $q \wedge \neg p$ and leads to $\mathbf{S}^{no}$. Only the states in $\mathbf{S}^{no}$ itself satisfy this property. So, $F = \{5\}$. $\mathbf{S}^{yes}$ (for ψ) is the complement of F . So, it is $\{0, 1, 2, 3, 4\}$.
- From the above we conclude that the probability of satisfying ψ is 1 on all states, except on state 5, for which the probability is 0.
- So, the states in $\{0, 1, 2, 3, 4\}$ satisfy $\mathbf{P}_{\geq 1}[\psi] = \chi$.

Next, we proceed to calculate the probabilities of different states to satisfy $p \mathbf{U} \chi$. From this we can see which states would satisfy $\mathbf{P}_{\leq 0.5} [p \mathbf{U} \chi]$ (which abbreviates $\mathbf{P}_{\leq 0.5} [p \mathbf{U} \mathbf{P}_{\geq 1}[q \mathbf{U} p]]$).

The calculation proceeds in the same way:

- We calculate the set E of states satisfying $\mathbf{E}[q \mathbf{U} \chi]$. Treat χ as an atomic proposition. Previously we already calculated which states satisfy χ , namely $\{0, 1, 2, 3, 4\}$; so, add χ as a 'label' to these states. We then obtain that E is also $\{0, 1, 2, 3, 4\}$. So, $\mathbf{S}^{no}$ (for $q \mathbf{U} \chi$) is $\{5\}$.
- Calculate the set F consisting states with an outgoing path whose states satisfy $q \wedge \neg \chi$ and leads to $\mathbf{S}^{no}$. Only the states in $\mathbf{S}^{no}$ itself satisfy this property. So, $F = \{5\}$. $\mathbf{S}^{yes}$ is the complement of F . So, it is $\{0, 1, 2, 3, 4\}$.
- From the above we conclude that the probability of satisfying $q \mathbf{U} \chi$ is 1 on all states, except on state 5, for which the probability is 0.
- So, the only state satisfying the asked property $\mathbf{P}_{\leq 0.5} [p \mathbf{U} \chi]$ is state 5.

6. PLTL formulas ϕ has the following syntax: where p is an atomic proposition, and $\sim$ is one of $\leq, <, \geq, >, =$.

$$\begin{aligned}\phi &::= \mathbf{P}_{\sim p}[\psi] \\ \psi &::= p \mid \psi_1 \wedge \psi_2 \mid \neg\psi \mid \mathbf{X}\psi \mid \psi_1 \mathbf{U} \psi_2 \quad (\text{path formulas})\end{aligned}$$

Note that a path formula can only recurse over other path formulas. In particular, unlike in PCTL, a PLTL path formula cannot have a probabilistic sub-formula.

Give the formal semantic of PLTL, interpreted over DTMC.

Answer: Let $M = (S, s_0, P, L)$ be your DTMC. A PLTL formula ϕ holds on M if ϕ holds on the initial state s_0 , denoted by $s_0 \models \phi$. In PLTL, ϕ takes the form of $\mathbf{P}_{\sim p}[\psi]$, for some path formula ψ . For any state s , the definition of $s \models \mathbf{P}_{\sim p}[\psi]$ is given below, for ' $>$ ' as the $\sim$. The definition is similar for other cases of $\sim$.

$$s \models \mathbf{P}_{>q}\psi = \underbrace{P\{\omega \in Paths(s) \mid \omega \models \psi\}}_{\text{probability}} > q$$

Here, $Paths(s)$ is the set of all possible paths/executions that starts from s ; typically there are infinitely many of them. The notation $\omega \models \psi$ means that the (temporal) property ψ holds on the path ω .

Finally, the notation $P\{\omega \in Paths(s) \mid \omega \models \psi\}$ refers to the probability that such a path ω occur as the behavior of the model if we start in the state s . This part is the same as what we had in the solution of question No. 1 on PCTL.

Since ψ is a purely LTL formula, the meaning of $\omega \models \psi$ is as standard in LTL. For your convenience this is shown below, using the structure as in Kwiatkowska-Parker (rather than the structure I used in the Lecture Notes):

$$\begin{aligned}\omega \models p &= p \in L(\omega_0) \\ \omega \models \neg\psi &= \text{not } \omega \models \psi \\ \omega \models \psi_1 \wedge \psi_2 &= \omega \models \psi_1 \text{ and } \omega \models \psi_2 \\ \omega \models \mathbf{X}\psi &= \omega[1..] \models \psi \\ \omega \models \psi_1 \mathbf{U} \psi_2 &= \omega \models \psi_2 \text{ or } (\omega \models \psi_1 \text{ and } \omega[1..] \models \psi_1 \mathbf{U} \psi_2)\end{aligned}$$

7. Given an example that demonstrates that PLTL formula $\mathbf{P}_{\geq 1}[F\psi]$ is not the same as plain LTL formula $\Diamond\psi$.

Answer: The same example as in No. 2. The PLTL $\mathbf{P}_{\geq 1}[Fp]$ holds on the DTMC there, but the plain LTL $\Diamond p$ is not valid on the model.

8. Consider again the DTMC M in No. 3.

- (a) What are the SCCs of M ? Which ones are bottom SCCs (BSCCs)?

Answer: There is only one SCC: the entire M . Therefore it is also a BSCC.

- (b) Does the PLTL property $\mathbf{P}_{\geq 1.0}[\mathbf{GF}(p \wedge q)]$ hold on M ?

Answer: Property of the form $\mathbf{GF}a$ where a is atomic can be checked directly via BSCCs. As mentioned above, the entire M is the one BSCC we have. Some states in M satisfies $p \wedge q$. So, $Prob(s, p, \mathbf{GF}(p \wedge q))$ from any state s in M is 1, including the initial state 0. In other words: yes, $\mathbf{P}_{\geq 1.0}[\mathbf{GF}(p \wedge q)]$ holds on M .

You might argue that within the BSCC M , the program could cycle e.g. $0 - 1 - 0 \dots$ forever, and hence avoiding $p \wedge q$. But note that the probability that this cycle occurs approaches 0. And similarly so for other cycles within the BSCC that avoids state 3 where $p \wedge q$ holds.

(c) How about $\mathbf{P}_{>0}[\mathbf{FG}p]$?

Answer: Property of the form $\mathbf{FG}a$ where a is atomic can also be checked directly via BSCCs. As said above, we only have one BSCC, which is the whole M . Not all states in this BSCC have p hold on them, e.g. p does not hold on state 5. So, $\text{Prob}(s, p, \mathbf{FG}p)$ from any state s in M , including the initial state 0, is zero. In other words: $\mathbf{P}_{>0}[\mathbf{FG}p]$ does *not* hold on M .

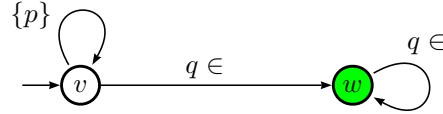
You might argue that within the BSCC M , the program could cycle $0 - 1 - 0 \dots$ forever, which has the property $\mathbf{FG}p$. But as pointed out before, the probability that this cycle occurs approaches 0.

9. Consider again the DTMC M in No. 3. We want to check if it satisfies the PLTL properties:

- $\mathbf{P}_{\leq 0.8}[(p \wedge \neg q) \mathbf{U} \mathbf{G}q]$.
- $\mathbf{P}_{\leq 0.8}[p \mathbf{U} \mathbf{G}q]$.

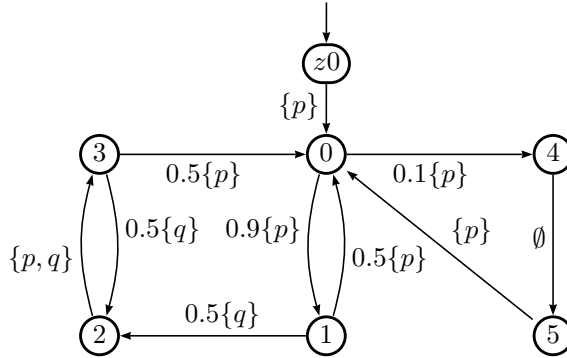
Perform the PLTL model checking algorithm to check them.

Answer: Let's first do property (a). The LTL property $\psi_a = (p \wedge \neg q) \mathbf{U} \mathbf{G}q$ can be expressed by the following deterministic Rabin automaton Ra :

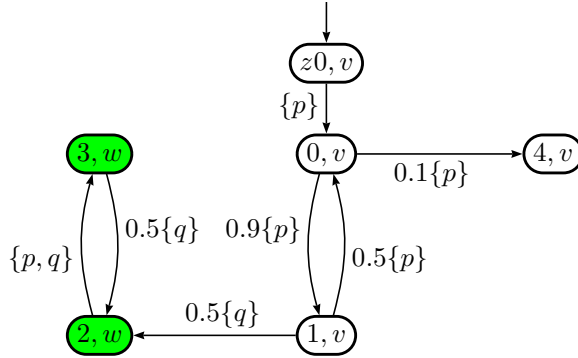


The accepting condition is $\{(L, K)\}$ where $L = \emptyset$ and $K = \{w\}$. Note: well, since L is empty, it is also a Buchi.

Before we construct the intersection $M \cap Ra$, let's first transform M by moving the labels to incoming arrows. We add a fake initial state too:



The intersection $M \cup Ra$ is shown below:



The accepting condition of $M \cup Ra$ is $\{(L, R)\}$ where L is empty (because the L of Ra was empty) and $K = \{(2, w), (3, w)\}$ (the green states).

The SCCs:

- $Z_0 = \{(z0, v)\}$
- $Z_1 = \{(4, v)\}$
- $Z_2 = \{(0, v), (1, v)\}$
- $Z_3 = \{(2, w), (3, w)\}$ (the green states)

Z_1 and Z_3 are BSCCs. Z_3 is an accepting BSCC because none of its members are in L (of $M \cup Ra$) and some of its members is in K . Z_1 is not an accepting BSCC.

Let $\psi_a = (p \wedge \neg q) \mathbf{U} \mathbf{G} q$. So, we have:

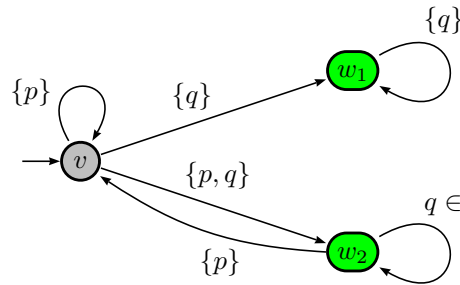
- From any state s in Z_3 , $Prob(s, \psi_a) = 1$.
- From any state s in Z_1 , $Prob(s, \psi_a) = 0$.
- $Prob((z0, v), \psi_a) = Prob((0, v), \psi_a)$.
- $Prob((0, v), \psi_a) = 0.1 * Prob((4, v), \psi_a) + 0.9 * Prob((1, v), \psi_a)$.
- $Prob((1, v), \psi_a) = 0.5 * Prob((0, v), \psi_a) + 0.5 * Prob((2, w), \psi_a)$.

Simplifying the equations give: $Prob((1, v), \psi_a) = 0.5 * 0.9 * Prob((1, v), \psi_a) + 0.5 * 1$. So, $Prob((1, v), \psi_a) = 50/55$.

So, $Prob((z0, v), \psi_a) = Prob((0, v), \psi_a) = 0.818....$

So, the PLTL property $\mathbf{P}_{\leq 0.8}[(p \wedge \neg q) \mathbf{U} \mathbf{G} q]$ does not hold on M .

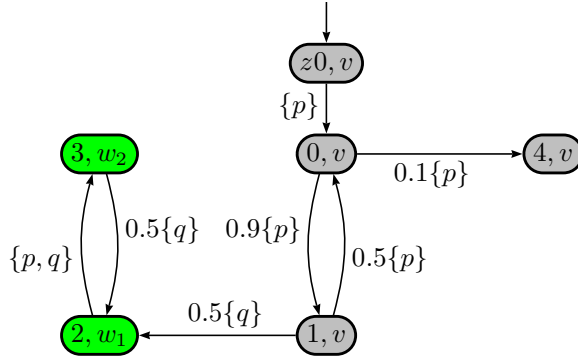
Ok, let's now check property (b). The LTL property $\psi_b = p \mathbf{U} \mathbf{G} q$ can be expressed by the following deterministic Rabin automaton Rb :



Accepting condition: $\{(L, K)\}$ where $L = \{v\}$ and $K = \{w1, w2\}$.

We now construct the intersection $M \cap Rb$. We end up with actually an automaton with the same structure as the previous $M \cap Ra$, though this is more a coincidence. The accepting condition is different though.

Note that going to w_2 is not possible because in L after transitions with $\{p\}$ we cannot immediately do a transition with $\{p, q\}$.



The accepting condition is $\{(L, K)\}$ where L consists of all states with v (so, the gray states) and K consists of all states with w_1 or w_2 (so, the green states).

Because the structure of the graph is the same as what we had with $M \cap Ra$, the SCCs and BSCCs are the same. We have one BSCC, containing the green states above. This BSCC is accepting, because it does not contain any state in L and it has at least one state from K .

Let $\psi_b = p \text{ U } \mathbf{G}q$. The probability $Prob((z0, v), psi_b)$ is the same of the probability of reaching the green-BSCC from the initial state $(z0, v)$. Because the intersection machine is the same as what we had with $M \cap Ra$ and the BSCC is the same, this probability would be just the same as $Prob((z0, v), psi_a) = 0.818...$

So, the PLTL property $\mathbf{P}_{\leq 0.8}[p ; \text{U } \mathbf{G}q]$ does not hold on M either.